Dr. Kevin Godin-Dubois

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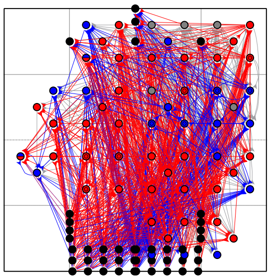
Position Researcher in Evolutionary Robotics (since November 2022)

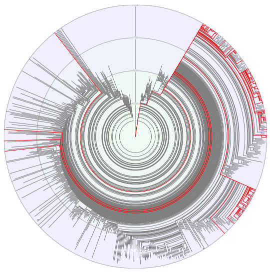
Highlights

Research *Artificial Life: Morphology, Cognition & Interaction*

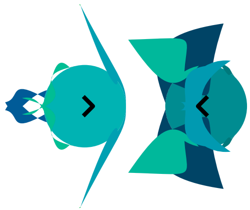
Main fields



Artificial Neural Networks



Species Dynamics



Morphogenetic Engineering

Output
3 journal articles (*Artificial Life, Frontiers, JOSS*)
5 international conference articles (*ALife, IEEE ALife, EvoAPP*)
7 international workshops short papers (*ALife, ECAL, AAAI*)
Scientific software: ABrain, AMaze, Revolve2, SHARPIE

Positions

Postdoctoral
2022 - Present
Computer Science / Evolutionary Robotics
"NeuroEvolution and Reinforcement Learning for Embodied Robots"
Computational Intelligence Group - Vrije Universiteit Amsterdam, The Netherlands
Supervisor: Dr. A. Yaman (a.yaman@vu.nl)
Collaborators: Dr. A. Kononova (a.kononova@liacs.leidenuniv.nl)

Postdoctoral
2020 - 2022
Computer Science / Artificial Intelligence
"Emergent cognitive architectures in virtual embodied robots"
REVA Team, IRIT - Toulouse I University, France
Supervisors: Pr. Y. Duthen (yves.duthen@irit.fr)
Pr. S. Cussat-Blanc (sylvain.cussat-blanc@irit.fr)

PhD
2016-2020
Computer Science / Artificial Life
"Environment-driven speciation: long term interactions in artificial plant communities"
REVA Team, IRIT - Toulouse I University, France
Supervisors: Pr. Y. Duthen (yves.duthen@irit.fr)
Pr. S. Cussat-Blanc (sylvain.cussat-blanc@irit.fr)

Teaching 9 years (∼800 hours)

Computer Science
Learning Machines, NeuroEvolution
Bachelor and Master research projects
Programming languages: Python, C, R
Algorithms, Data Structures, Information theory
Programming projects

Generalists
Data Science tools and languages
Database modeling, SQL

Skills

Programming
Fluent: C++, Bash, Python, LaTeX
Working Knowledge: C, Java, R, VB, VBA, JavaScript
Frameworks: (Py)Qt, MuJoCo

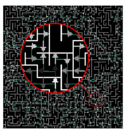
Technical
Evolutionary Algorithms, Machine Learning, Multi-Agents Systems, High-Performance Computing

Languages
French (native), English

Research

Synopsis
My main interests revolve around autonomous artificial life forms: from the design of efficient morphologies to the emergence of high-level control schemes and the evolutionary constraints that favor both. Recently I am mostly focused on Artificial Neural Networks (ANN) through NeuroEvolution and Reinforcement Learning, notably in the context of Interactive Evolutionary Robotics.

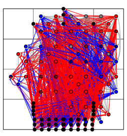
Reinforcement Learning
Complementing Neuro-Evolutionary approaches with Reinforcement-based learning for more robust agents.



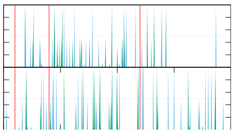
[1, 2] AMAZE
Benchmark generator for lightweight sighted agents. Meant as a fast prototyping platform for Reinforcement Learning and NeuroEvolutionary algorithms.
Software: AMaze

[13, 12] SHARPIE
Collaborative Hybrid Intelligence project aiming to provide the new standard for interactive experiments in Reinforcement Learning settings.
Software: SHARPIE

Artificial Neural Networks
Studying the emergence of various "cognitive" capabilities in virtual robots, controlled by a spontaneously differentiated neural network, in response to biologically plausible stimuli.



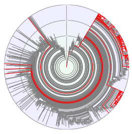
[8, 3, 7] VIRTUAL FMRI
Extracting stimulus-specific regions of an ANN by applying a virtual equivalent to functional Magnetic Resonance Imaging (fMRI) and building high-level cognitive maps.
Software: ABrain



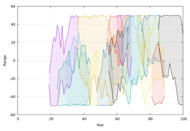
[16, 15] Communication
Exploring the mechanisms leading to emergent communication, how it becomes structured and its neural implementation.

Species Dynamics

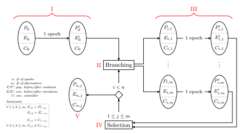
Promoting complex evolutionary trajectories and extracting species-level information from individual reproductions.



[19, 17] Phylogenetics
Automatically transforming genealogic trees into phylogenetic abstraction to access the emergent species-level dynamics.
Software: APOGeT(Automated Phylogeny Over Geological Timescales)



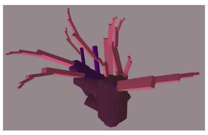
[11] Speciation
Application of a bio-inspired reproduction operator (Bail-Out Crossover) capable of spontaneously generating species barriers thereby allowing for emergent speciation.



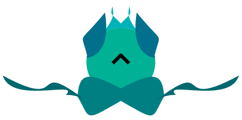
[9, 24] Evolutionary algorithms
Introduced a novel paradigm, EDEnS (Environment-Driven Evolutionary Selection), relying on the indirect controlling of whole populations' evolutionary trajectories through an evolvable environmental controller.

Morphogenetic Engineering

Developing functional morphology in response to environmental constraints and evolutionary pressures.



[18, 9, 10, 19] Developmental morphologies
Production of mature, functional virtual plants from a single cell/structure using various genetic encodings (rules-based, L-Systems, Graphtals) in response to environmental constraints.



[3] Virtual robots
Use of genetically parameterized cubic bézier curves to control both static and mobile structures on the perimeter of virtual circular robots.
Software: Splinoids **Videos:** on Vimeo

Expertise

Evolutionary Algorithms
- Environment-Driven Evolutionary Selection (EDEnS)
- Multi-objective Optimisation
- High Performance Computing (HPC), Co-evolution, Novelty

Machine Learning
- Artificial Neural Networks (ANN, CNN, RNN)
- Composite Pattern-Producing Networks (CPPN)
- Cartesian Genetic Programming (CGP)
- Genetic Regulatory Networks (GRN)
- Hidden Markov Models (HMM)
- Stable baselines 3, PyTorch

Physics simulators
- Approximate: Bullet3, Box2D
- Exact: MuJoCo

Teachings

Postdoc 2023-2024	Vrije Universiteit Amsterdam	
	• NeuroEvolution (lecture)	
	• Learning Machines (projects)	*45h*
	• Master and Bachelor thesis supervision	*~200h*
Course management 2021-2022	Toulouse I University & Toulouse III University, France	
	• Computer Science projects	*72h*
	Multi-Agent Systems, Complex Systems, Simulation	
	• R programming	*67.5h*
	English lectures	
	• Information theory	*22.5h*
	• Servers and contents	*18.75h*
Teaching fellow 2017-2021	Toulouse I University & Toulouse III University, France	
	• Statistical software (R & Python)	*36h*
	• Algorithms	*60h*
	• Excel & VBA	*60h*
	• Modeling in databases	*21h*
Practical work supervisor 2016-2021	Toulouse III University, France	
	• Software projects	*69.2h*
	• Data structures	*18.8h*
	• C Programming	*36h*
	• Python	*8h*

Outreach

Reviewer
Since 2023

- Artificial Life (ALife) program comitee member
- IEEE Transactions on Evolutionary Computation
- Journal of Open Source Software reviewer
- PPSN 2026

EduMix Aspi-Friendly
2021

Initiated a project for the self-monitoring of well-being in students with autistic disorders alongside a heterogeneous team of neuro-(a)typical and various profiles (faculty, designers, developers ...).

Internships

Morphogenetic Engineering
2016 (6 months)

Toulouse Research Institute on Computer Science (IRIT), France
"Rule-based artificial embryogenesis in a complex 3D environment"
Deployed rule-based genomes on the MecaCell platform to study artificial plant growth and cell specialization.
Contact: Pr. Y. Duthen (`yves.duthen@irit.fr`)

Machine Learning 2015 (3 months)	IRIT, *"Comparison of different evolutionary approaches, an application to the GECCO 2015 challenge"* Performed a performance comparison (accuracy, efficiency) between Artificial Neural and Genetic Regulatory Networks on the 2015 GECCO temperature prediction challenge data. **Contact:** Pr. H. Luga (`herve.luga@irit.fr`)
Machine Learning 2014 (2 months)	IRIT, *"An architecture for automated bird discrimination"* Applied Hidden Markov Models to the BirdClef2014 challenge on the identification of specific bird species in a corpus of thousands of recordings. **Contact:** Pr. J. Farinas (`jerome.farinas@irit.fr`)

Education

PhD 2016 - 2020	Toulouse I University, France Defended the 15th of July 2020 Thesis title: *"Environment-driven speciation: long term interactions in artificial plant communities"* Investigated how complexification of artificial creatures could be further enhanced through the indirect control provided by a co-evolved, highly dynamical environment. **Rapporteurs:** Pr. P. Collet & DoR. F. Vico **Contact:** Pr. Y. Duthen (`yves.duthen@irit.fr`)
Master 2014 - 2016	Toulouse III University, France (*with honours*) Artificial Intelligence: mathematical & symbolic models, training methods
Bachelor 2011 - 2014	Toulouse III University (*with distinction*) Computer Science: networks, programming, systems, mathematics

Scholarships and Fellowships

2023-2026 ∼ 200K €	Postdoctoral funding from the Hybrid Intelligence consortium (Netherlands)
2016-2019 70K €	PhD Fellowship from the French Minister of Higher Education and Research (MESR)
2015 10K €	Master Scholarship from the International Mathematics and Computer Science Center (LabEx CIMI, Toulouse)
2014-2015 3K6 €	Merit Scholarship from the Regional Student Welfare Office (CROUS, Toulouse)

Research Output

Journals (peer-reviewed)

[1] **K. Godin-Dubois**, K. Miras, and Anna V Kononova. "AMaze: An Intuitive Benchmark Generator for Fast Prototyping of Generalizable Agents". In: *Frontiers in Artificial Intelligence* Volume 8 - 2025 (2025). ISSN: 2624-8212. DOI: 10.3389/frai.2025.1511712.

[2] **K. Godin-Dubois**, K. Miras, and A. V. Kononova. "AMaze: A Benchmark Generator for Sighted Maze-Navigating Agents". In: *Journal of Open Source Software* (2025). DOI: 10.21105/joss.07208.

[3] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. "Explaining the Neuroevolution of Fighting Creatures Through Virtual fMRI". In: *Artificial Life* 29.1 (2023), pp. 66–93. ISSN: 1064-5462. DOI: 10.1162/artl_a_00389.

International conferences (peer-reviewed)

[4] **K. Godin-Dubois**, A. Yaman, and A. V. Kononova. "Benefits of Low-Cost Bio-Inspiration in the Age of Overparametrization". In: *Artificial Evolution*. 19. Nice, France: arXiv, 2026, under review. DOI: 10.48550/ARXIV.2604.20365. (Visited on 04/23/2026).

[5] L. Goncalvez Braz, F. Den Hengst, **K. Godin-Dubois**, H. Aydin, K. Baraka, S. Wang, and F. A. Oliehoek. "SHARPIE: A Robust Framework for Streamlining Human-AI Reinforcement Learning Experiments". In: *European Workshop on Reinforcement Learning*. 2026, in press.

[6] O. Moulin, D. Pasero, L. Bierling, and **K. Godin-Dubois**. "On the Impact of Morphological Encoding for Evolutionary Robotics in ARIEL". In: *Artificial Evolution*. 19. Nice, France, 2026, under review.

[7] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. "Specialization or Generalization: Investigating NeuroEvolutionary Choices via Virtual fMRI". In: *ALIFE 2024: Proceedings of the 2024 Artificial Life Conference*. MIT Press, July 2024. DOI: 10.1162/isal_a_00817.

[8] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. "Spontaneous Modular NeuroEvolution Arising from a Life/Dinner Paradox". In: *The 2021 Conference on Artificial Life*. Cambridge, MA: MIT Press, 2021, p. 95. DOI: 10.1162/isal_a_00431. Presentation: https://vimeo.com/godinduboisalife/alife2021main.

[9] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. "Beneficial Catastrophes: Leveraging Abiotic Constraints through Environment-Driven Evolutionary Selection". In: *2020 IEEE Symposium Series on Computational Intelligence (SSCI)*. 2020, pp. 94–101. DOI: 10.1109/SSCI47803.2020.9308411.

[10] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. "Self-Sustainability Challenges of Plants Colonization Strategies in Virtual 3D Environments". In: *Applications of Evolutionary Computation*. Ed. by P. Kaufmann and P. A. Castillo. Cham: Springer International Publishing, 2019, pp. 377–392. ISBN: 978-3-030-16692-2. DOI: 10.1007/978-3-030-16692-2_25.

[11] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. "Speciation under Changing Environments". In: *ALIFE 19*. Vol. 31. Cambridge, MA: MIT Press, 2019, pp. 349–356. ISBN: 978-0-262-35844-6. DOI: 10.1162/isal_a_00186. Presentation: https://vimeo.com/godinduboisalife/alife2019.

Workshops

[12] H. Aydin, **K. Godin-Dubois**, L. G. Braz, F. Den Hengst, K. Baraka, M. M. Celikok, A. Sauter, S. Wang, and F. A. Oliehoek. “SHARPIE: A Modular Framework for Reinforcement Learning and Human-AI Interaction Experiments”. In: *Collaborative AI and Modeling of Humans*. arXiv, Feb. 2025. DOI: `10.48550/arXiv.2501.19245`. (Visited on 03/14/2025).

[13] H. Aydin, **K. Godin-Dubois**, L. Goncalvez Braz, Floris Den Hengst, K. Baraka, M. M. c Celikok, A. Sauter, S. Wang, and F. A. Oliehoek. “Towards an Experimentation Platform for Hybrid Human-AI Sequential Decision-Making”. In: *Frontiers in Artificial Intelligence and Applications*. Ed. by D. Pedreschi, M. Milano, I. Tiddi, S. Russell, C. Boldrini, L. Pappalardo, A. Passerini, and S. Wang. IOS Press, Sept. 2025. ISBN: 978-1-64368-611-0. DOI: `10.3233/FAIA250669`.

[14] **K. Godin-Dubois**, O. Weissl, K. Miras, and A. V. Kononova. “Interactive Embodied Evolution for Socially Adept Artificial General Creatures”. In: *Evolution of Things Workshop at the ALife 2024 Conference*. arXiv, July 2024. DOI: `10.48550/arXiv.2407.21357`.

[15] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “Emergent Communication for Coordination in Teams of Embodied Agents”. In: *4th International Workshop on Agent-Based Modelling of Human Behaviour (ALife2022)*. 2022.

[16] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “On the Benefits of Emergent Communication for Threat Appraisal”. In: *3rd International Workshop on Agent-Based Modelling of Human Behaviour*. Online, 2021. Presentation: `https://vimeo.com/godinduboisalife/abmhub2021`.

[17] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “APOGeT: Automated Phylogeny Over Geological Timescales”. In: *MethAL Workshop at ALife 2019*. 2019. DOI: `10.48550/arXiv.2407.21412`.

[18] **K. Dubois**, S. Cussat-Blanc, and Y. Duthen. “Towards an Artificial Polytrophic Ecosystem”. In: *Morphogenetic Engineering Workshop, at the European Conference on Artificial Life (ECAL) 2017 September 4*. 2017.

Posters

[19] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. “Studying Long Term Interactions between Plants and Their Environment”. In: *Alife 2018*. Tokyo, 2018. DOI: `10.13140/RG.2.2.27553.97125`.

Oral presentations

[20] **K. Godin-Dubois**, S. Cussat-Blanc, and Y. Duthen. *Splinoids: First Steps out of EDEnS*. Talk. Montreal (Virtual), 2020.

Softwares and datasets

[21] **K. Godin-Dubois**. *AMaze: Fully Discrete Training with Three Regimes (Direct, Scaffolding, Interactive) and Two Algorithms (A2C, PPO)*. Dataset. Feb. 2024. DOI: `10.5281/ZENODO.10622913`. (Visited on 09/19/2024).

[22] **K. Godin-Dubois**, K. Miras, and A. V. Kononova. *AMaze: A Lightweight Benchmark Generator for Sighted Agents*. Zenodo. Software. Apr. 2024. DOI: `10.5281/ZENODO.10907939`.

[23] A. Stuurman, O. Weissl, T.-C. Chiang, AndresG, D. Zeeuwe, **K. Godin-Dubois**, and Roy. *Ci-Group/Revolve2: 1.2.3*. Zenodo. Software. Nov. 2024. DOI: 10.5281/ZENODO.14143431. (Visited on 02/12/2025).

Thesis

[24] **K. Godin-Dubois**. "Environment-Driven Speciation: Long-Term Interactions in Artificial Plant Communities". PhD thesis. Doctoral school of Mathematics, Computer Science and Telecommunications (Toulouse, France), 2020.